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## ► To cite this version:

Karthika Nasir, Rahul Menon, Sneha Iyer. A Mathematical and Algorithmic Review of Physics-Informed Neural Networks for Scientific Computing. 2025. hal-05053836

**HAL Id: hal-05053836**

**<https://hal.science/hal-05053836v1>**

Preprint submitted on 2 May 2025

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# A Mathematical and Algorithmic Review of Physics-Informed Neural Networks for Scientific Computing

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**Abstract.** In recent years, the advent of *Physics-Informed Neural Networks* (PINNs) has inaugurated a novel computational paradigm wherein the governing laws of physical systems—typically encoded as partial differential equations (PDEs), ordinary differential equations (ODEs), or integral-differential operators—are embedded directly into the training objective of deep neural networks. This embedding yields a hybrid model that seeks to approximate the underlying solution manifold by minimizing a composite loss functional involving both observational data and analytically prescribed physical constraints. The resulting framework offers a promising route toward data-efficient, mesh-free, and flexible solvers for a broad class of forward, inverse, and hybrid problems arising in fluid dynamics, solid mechanics, electromagnetism, quantum systems, and beyond. However, this promise is tempered by numerous theoretical, algorithmic, and practical challenges that remain unresolved.

This survey endeavors to provide an exhaustive and critical account of the mathematical foundations, algorithmic formulations, architectural design principles, and representative applications of PINNs, while simultaneously delineating the unresolved issues that continue to delimit their efficacy and generalizability. We begin by reviewing the fundamental construction of PINNs, including their variational formulation, collocation-based enforcement of PDE constraints, and coupling with observational data through empirical risk minimization. We analyze the interplay between network expressivity and PDE regularity, elucidate the role of automatic differentiation in gradient computation, and examine recent advances such as adaptive sampling, domain decomposition (XPINNs), and operator learning. Subsequently, we scrutinize the empirical successes of PINNs across canonical benchmarks and real-world systems, highlighting their ability to learn latent dynamics, infer unknown constitutive laws, and reconstruct unmeasured states under limited supervision.

Despite these advances, PINNs face several fundamental limitations, including optimization instability due to non-convex loss landscapes with high-order derivatives, poor generalization under spectral bias, difficulty in accurately resolving multiscale and stiff systems, and inadequate enforcement of boundary conditions in complex geometries. The lack of

theoretical guarantees regarding convergence, stability, and error bounds further complicates their interpretation and deployment in mission-critical contexts. In response, we explore alternative formulations, such as variational PINNs, probabilistic PINNs, and neural operators, and identify directions for future research, including the development of physics-adaptive architectures, scalable and PDE-aware optimization schemes, and formal mathematical theories grounded in functional analysis and information theory.

In conclusion, we argue that the continued evolution of PINNs will require a convergence of insights from numerical analysis, applied mathematics, statistical learning theory, and physics-based modeling. This synthesis holds the potential to not merely advance the state of computational science, but to redefine our conceptual framework for integrating physical laws and empirical data within a unified learning-theoretic formalism.

**Keywords:** Physics-Informed Neural Networks (PINNs); Partial Differential Equations (PDEs); Scientific Machine Learning; Deep Learning; Operator Learning; Inverse Problems; Data-Driven Modeling; Uncertainty Quantification; Numerical Methods; Function Approximation Theory.

## 1 Introduction

In recent years, the rapidly evolving interface between computational physics and artificial intelligence has engendered a class of methodologies collectively referred to as Physics-Informed Neural Networks (PINNs), wherein the a priori encoded structure of physical laws—frequently expressed as systems of partial differential equations (PDEs), integral constraints, and conservation identities—is inherently imposed upon the architecture and loss functionals of deep neural networks [1]. This integration represents a paradigmatic shift from purely data-driven machine learning to a hybridized framework where empirical observations and deterministic theoretical constructs are interwoven with algorithmic approximation schemes, allowing for solution representations that are not only data-consistent but also intrinsically constrained by governing physical principles. The formalism of PINNs, as initially articulated in seminal works such as those by Raissi, Perdikaris, and Karniadakis, is premised on the minimization of a composite loss function  $\mathcal{L}$  that encodes both data fidelity and residual minimization of the PDEs. Letting  $\mathcal{N}_\theta$  denote a neural network parameterized by weights and biases  $\theta$ , and given a PDE of the general form

$$\mathcal{D}[u(\mathbf{x}, t)] = f(\mathbf{x}, t), \quad (1)$$

where  $\mathcal{D}[\cdot]$  is a differential operator and  $f$  is a known source term, the PINN methodology involves the minimization of the loss

$$\mathcal{L}(\theta) = \lambda_d \mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{physics}}, \quad (2)$$

where  $\mathcal{L}_{\text{data}}$  quantifies discrepancy between  $\mathcal{N}_\theta$  and empirical data, while  $\mathcal{L}_{\text{physics}}$  penalizes deviations from the PDE residual  $\mathcal{D}[\mathcal{N}_\theta] - f$  evaluated at collocation points [2]. The scalar weights  $\lambda_d$  and  $\lambda_p$  are hyperparameters regulating the trade-off between data adherence and physical consistency. This mathematical synthesis enables the surrogate modeling of high-dimensional, nonlinear, and even partially observed physical systems, wherein the model architecture is neither purely data interpolative nor purely simulation-based [3]. Rather, it embodies a constrained optimization problem in function space, where the solution manifold is implicitly regularized by physical knowledge [4]. Such an approach possesses significant ramifications across domains characterized by sparse data regimes, high computational costs of traditional solvers, and the necessity for robust generalization beyond interpolation domains [5]. Despite its conceptual elegance and empirical efficacy, the PINN framework remains beset by multifarious challenges [6]. These include, but are not limited to, issues of convergence stagnation in stiff or chaotic regimes, gradient pathologies arising from non-convex optimization landscapes, hyperparameter sensitivity, and difficulties in enforcing hard constraints for systems with discontinuities or multi-scale features [7, 8]. Moreover, the theoretical underpinnings of generalization error, approximation capacity, and expressiveness in the context of physically structured function spaces remain areas of active inquiry [9]. In this survey, we endeavor to systematically dissect the mathematical structure, algorithmic strategies, and representative applications of Physics-Informed Neural Networks. Our exegesis spans recent theoretical advances, methodological innovations (such as adaptive sampling, transfer learning, and adversarial formulations), and diverse application frontiers, ranging from fluid mechanics and quantum systems to inverse problems in medical imaging and geophysics [10]. Furthermore, we elucidate the prevailing limitations and propose prospective avenues of research that may bridge the current epistemic gaps and propel the field toward greater theoretical rigor and practical ubiquity.

## 2 Mathematical Formulation and Algorithmic Foundations

The theoretical edifice upon which the paradigm of Physics-Informed Neural Networks (PINNs) is constructed rests upon a confluence of functional analysis, variational calculus, approximation theory, and modern machine learning [11]. At its core, a PINN may be interpreted as a solution ansatz  $u_\theta(\mathbf{x}, t)$  that resides in a parametric function space  $\mathcal{F}_\Theta = \{\mathcal{N}_\theta : \theta \in \Theta \subseteq \mathbb{R}^P\}$ , where  $\mathcal{N}_\theta$  denotes a neural network with parameters  $\theta$ , mapping input coordinates  $(\mathbf{x}, t) \in \Omega \times [0, T]$  to target variables  $u \in \mathbb{R}^m$  [12]. The overarching objective is to approximate the true solution  $u(\mathbf{x}, t)$  of a governing physical system, subject to partial differential equations (PDEs), boundary conditions, and initial constraints, through minimization of an appropriate loss functional defined over the space  $\mathcal{F}_\Theta$ . Let us

consider, without loss of generality, a canonical form of a time-dependent PDE:

$$\mathcal{D}[u(\mathbf{x}, t)] \equiv \frac{\partial u}{\partial t} + \mathcal{N}_{\text{spatial}}[u(\mathbf{x}, t)] = f(\mathbf{x}, t), \quad (\mathbf{x}, t) \in \Omega \times [0, T], \quad (3)$$

where  $\mathcal{N}_{\text{spatial}}$  is a nonlinear spatial differential operator (possibly involving derivatives of various orders, nonlinear couplings, or integral terms), and  $f$  denotes an external forcing term [13]. This equation is supplemented by initial conditions of the form

$$u(\mathbf{x}, 0) = u_0(\mathbf{x}), \quad \mathbf{x} \in \Omega, \quad (4)$$

and boundary conditions which may be of Dirichlet, Neumann, or Robin type, defined over the spatial boundary  $\partial\Omega$  [14]. In the PINN framework, a neural network  $\mathcal{N}_\theta$  is tasked with approximating the unknown solution  $u$  [15]. The fidelity of this approximation is measured through the minimization of a composite loss function  $\mathcal{L}_{\text{PINN}}$ , typically structured as:

$$\mathcal{L}_{\text{PINN}}(\theta) = \lambda_{\text{data}}\mathcal{L}_{\text{data}} + \lambda_{\text{PDE}}\mathcal{L}_{\text{res}} + \lambda_{\text{IC}}\mathcal{L}_{\text{init}} + \lambda_{\text{BC}}\mathcal{L}_{\text{bc}}, \quad (5)$$

where the constituent terms are defined as follows:

- $\mathcal{L}_{\text{data}} = \frac{1}{N_d} \sum_{i=1}^{N_d} \|\mathcal{N}_\theta(\mathbf{x}_i^d, t_i^d) - u_i^d\|^2$  penalizes discrepancies between network outputs and observed data points [16].
- $\mathcal{L}_{\text{res}} = \frac{1}{N_r} \sum_{i=1}^{N_r} \|\mathcal{D}[\mathcal{N}_\theta](\mathbf{x}_i^r, t_i^r) - f(\mathbf{x}_i^r, t_i^r)\|^2$  imposes PDE residual constraints at collocation points [17].
- $\mathcal{L}_{\text{init}} = \frac{1}{N_0} \sum_{i=1}^{N_0} \|\mathcal{N}_\theta(\mathbf{x}_i^0, 0) - u_0(\mathbf{x}_i^0)\|^2$  enforces initial condition compliance [18].
- $\mathcal{L}_{\text{bc}}$  accounts for boundary condition enforcement, the formulation of which varies depending on the boundary type [19].

The differential operators in  $\mathcal{L}_{\text{res}}$  are computed via automatic differentiation, a procedure in which the computational graph of the neural network is leveraged to obtain exact (machine-precision) derivatives without numerical discretization. This exactness, however, is bounded by machine precision and can become ill-conditioned in stiff regimes or for high-order derivatives [20]. A noteworthy aspect of PINNs is that they are mesh-free by design, eschewing the requirement of structured spatial discretization as in finite difference or finite element methods [21]. Instead, collocation points  $\{(\mathbf{x}_i^r, t_i^r)\}_{i=1}^{N_r}$  are sampled from the domain  $\Omega \times [0, T]$  using uniform grids, Latin hypercube sampling, or adaptive refinement schemes that concentrate points in regions of high residual error [10]. This renders PINNs particularly suited to problems involving irregular geometries or moving boundaries, wherein mesh generation is prohibitively expensive or algorithmically cumbersome [22].

The optimization landscape of  $\mathcal{L}_{\text{PINN}}(\theta)$  is typically nonconvex, non-smooth, and sensitive to the initialization of  $\theta$  as well as to the magnitudes of the weighting coefficients  $\lambda_i$  [24]. Empirical studies have shown that improper balancing of these weights can lead to underfitting of physical constraints or overfitting

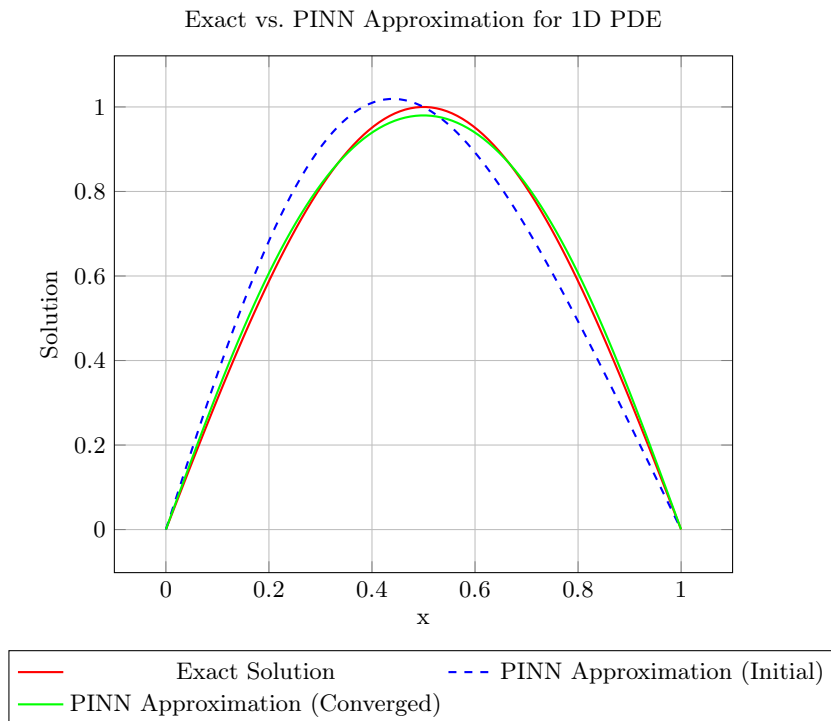


Fig. 1: Comparison between the exact solution and PINN approximations for a 1D PDE [23]. The initial PINN approximation (dashed blue curve) shows a noticeable error, but as the network converges, the approximation (green curve) becomes closer to the exact solution (red curve).

to data, suggesting the necessity of adaptive weighting strategies [25]. Techniques such as dynamic weighting, residual-based adaptive resampling (RAR), curriculum learning, and Sobolev training have been proposed to ameliorate such issues. Moreover, in certain contexts, the function  $u$  may exhibit multiple time or spatial scales, or manifest non-smooth behavior such as shocks, discontinuities, or topological transitions [26]. Traditional PINNs, relying on global smooth approximations via differentiable neural networks, struggle to capture such localized phenomena [27]. Remedies to this include the use of domain decomposition (as in XPINNs), the adoption of local neural networks (multi-network PINNs), and the incorporation of solution-informed basis expansions [28]. From a theoretical perspective, the approximation capacity of neural networks in the PINN framework can be understood through the lens of Sobolev space embeddings and universal approximation theorems in Banach spaces. If  $\mathcal{N}_\theta$  is capable of approximating functions in a Sobolev space  $W^{k,p}(\Omega)$ , then under suitable regularity conditions, it can approximate the solution  $u$  with error bounded in the Sobolev norm. However, explicit error bounds and convergence rates for PINNs remain

largely empirical or heuristic, especially in high-dimensional settings, where the curse of dimensionality impairs sampling efficiency and optimization tractability [29]. Finally, the extension of PINNs to stochastic systems, parameterized PDEs, and inverse problems introduces additional complexity [30]. In such cases, the network is trained not only to recover  $u$ , but also to infer hidden parameters  $\mu$  or latent variables. The resulting loss functions become nested optimization problems, sometimes bi-level in structure, and often necessitate Bayesian reformulations, variational inference, or physics-constrained generative models [31]. In summation, the mathematical and algorithmic structure of PINNs constitutes a highly flexible, albeit intricate, framework for physics-constrained learning [32]. Its success is predicated on a careful orchestration of network architecture, optimization dynamics, sampling strategy, and theoretical understanding. In the subsequent sections, we shall examine specific applications and extensions of PINNs, elucidating their performance across domains as diverse as turbulent fluid dynamics, quantum mechanical systems, and biomedical imaging, while continuing to explore their limitations and possible resolutions [33].

### 3 Applications Across Physical Domains

The versatility of Physics-Informed Neural Networks (PINNs) is manifest in their capacity to transcend disciplinary boundaries and operate as general-purpose solvers, emulators, and inference engines for systems governed by partial differential equations (PDEs) and associated variational principles. Their formulation, rooted in universal function approximation theorems yet enriched by domain-specific physical priors, allows for their deployment across a panoply of physical sciences [34]. In this section, we undertake a detailed exposition of PINN applications across several canonical domains, including but not limited to fluid dynamics, quantum mechanics, solid mechanics, electromagnetism, geophysics, and biomedical imaging. For each, we highlight the unique characteristics of the underlying mathematical models, elucidate the corresponding PINN formulations, and critically assess empirical performance and domain-specific challenges.

#### 3.1 Fluid Dynamics

The domain of fluid dynamics provides a particularly fertile ground for the application of PINNs, given the ubiquity and complexity of the Navier-Stokes equations [35]. These equations, which describe the momentum and mass conservation in viscous incompressible flows, take the form:

$$\nabla \cdot \mathbf{u} = 0, \quad (6)$$

$$\frac{\partial \mathbf{u}}{\partial t} + (\mathbf{u} \cdot \nabla) \mathbf{u} = -\nabla p + \nu \Delta \mathbf{u}, \quad (7)$$

where  $\mathbf{u}(\mathbf{x}, t)$  denotes the velocity field,  $p(\mathbf{x}, t)$  the pressure field, and  $\nu$  the kinematic viscosity. The nonlinearity and potential stiffness of these equations pose

substantial challenges to classical numerical methods, particularly in regimes of high Reynolds number or near turbulence onset. In the PINN framework, a neural network  $\mathcal{N}_\theta : \mathbb{R}^{d+1} \rightarrow \mathbb{R}^{d+1}$  is typically trained to map spatial-temporal inputs  $(\mathbf{x}, t)$  to  $(\mathbf{u}, p)$  outputs, with the residuals of Eqs. (6) and (7) forming the core of the physics loss [15]. Notably, enforcing incompressibility  $\nabla \cdot \mathbf{u} = 0$  often necessitates careful architectural design, such as divergence-free network parametrizations, or the imposition of this constraint via soft or hard penalties. Applications range from low-dimensional benchmark flows—such as the lid-driven cavity, flow past a cylinder, and Taylor-Green vortices—to more sophisticated three-dimensional simulations involving vortex shedding and transitional turbulence [36]. While PINNs have demonstrated the capacity to accurately reconstruct flow fields from sparse or noisy measurements, they often suffer from long training times, and difficulties capturing sharp gradients or near-wall behavior, thereby motivating hybrid approaches involving PINNs and traditional solvers.

### 3.2 Quantum Mechanics

In quantum mechanics, the Schrödinger equation governs the temporal and spatial evolution of quantum states:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \Delta \psi + V(\mathbf{x})\psi, \quad (8)$$

where  $\psi(\mathbf{x}, t)$  is the complex-valued wavefunction,  $V(\mathbf{x})$  the potential, and  $m$  the particle mass. The intrinsic linearity and spectral properties of this equation render it amenable to PINN-based solution strategies, provided that complex-valued function approximators are employed, or that real and imaginary components are treated separately. PINNs have been successfully used for solving both time-dependent and time-independent Schrödinger equations, including eigenvalue problems where the goal is to identify stationary states and corresponding energy levels. In inverse quantum problems, such as inferring the potential  $V(\mathbf{x})$  from partial observations of  $\psi$ , PINNs serve as a bridge between sparse experimental data and latent Hamiltonian structure, often outperforming black-box regression models in physical consistency and extrapolation capacity [37].

### 3.3 Solid Mechanics

In the realm of elasticity and solid mechanics, the governing equations are typically derived from the balance of linear momentum and strain-stress relations [38]. For a linear elastic isotropic medium, the Navier-Cauchy equations are given by:

$$\mu \Delta \mathbf{u} + (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}) + \mathbf{f} = \rho \frac{\partial^2 \mathbf{u}}{\partial t^2}, \quad (9)$$

where  $\mathbf{u}$  is the displacement field,  $\lambda$  and  $\mu$  are Lamé parameters, and  $\mathbf{f}$  represents body forces. The inherent difficulty of modeling complex material geometries,

interfaces, and loading conditions has motivated the application of PINNs to problems involving fracture mechanics, heterogeneous media, and stress field estimation from sparse sensor data [39]. Particularly in inverse mechanics, where one aims to recover material properties or loading patterns from partial displacement observations, PINNs offer a principled framework for integrating prior constitutive laws with empirical observations.

### 3.4 Electromagnetism

Maxwell’s equations form the backbone of classical electrodynamics:

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0}, \quad \nabla \cdot \mathbf{B} = 0, \quad (10)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t}. \quad (11)$$

Here,  $\mathbf{E}$  and  $\mathbf{B}$  are the electric and magnetic fields, respectively,  $\rho$  is the charge density, and  $\mathbf{J}$  the current density [40]. These equations pose both analytical and computational challenges when applied to heterogeneous media, time-varying boundary conditions, or wave propagation in complex domains. PINNs offer a mesh-free and constraint-satisfying alternative for solving Maxwell’s equations, with applications spanning antenna design, photonic crystals, and metamaterials [41]. The enforcement of divergence constraints, however, necessitates either penalty terms or the use of compatible numerical schemes that respect the Gauss laws.

### 3.5 Geophysics and Biomedical Imaging

In geophysics, the subsurface inference from sparse surface data is often modeled via elliptic or parabolic PDEs, such as the diffusion or Helmholtz equations [42]. PINNs have been applied to problems in seismic tomography, where the wave speed or subsurface structure is inferred from scattered wavefield data [43]. Their ability to integrate measurement noise with physical wave propagation constraints allows for improved robustness over conventional inversion algorithms [44]. Similarly, in biomedical imaging, such as electrical impedance tomography (EIT) or magnetic resonance elastography (MRE), the underlying problems are often ill-posed inverse PDE systems [45]. PINNs are deployed to enforce regularization via known physical laws, often achieving improved reconstructions under conditions of limited or noisy data [46].

### 3.6 Discussion and Emerging Domains

The efficacy of PINNs across such disparate fields illustrates their fundamental utility as generalized PDE solvers and physics-constrained inference mechanisms. Nonetheless, the aforementioned applications also underscore their limitations: sensitivity to problem scaling, training instabilities in stiff systems, and performance degradation in high-dimensional or noisy regimes. Efforts to address

these challenges include architectural modifications (Fourier-feature networks, SIRENs), algorithmic innovations (residual minimization scheduling, adaptive sampling), and hybrid schemes incorporating classical solvers into the training loop.

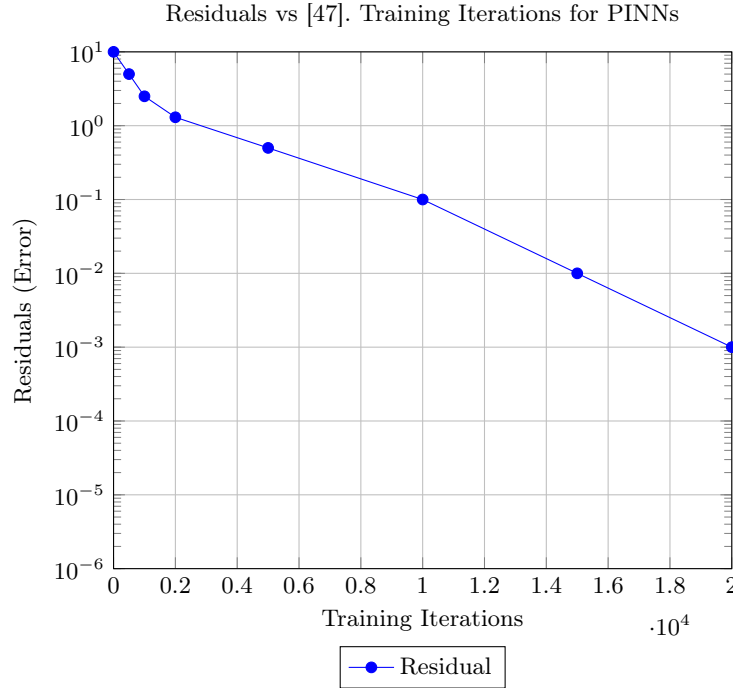


Fig. 2: Plot of residuals (error) as a function of training iterations for a PINN optimization process. The residuals decrease over time as the network trains, following a typical convergence behavior.

Emerging domains such as relativistic hydrodynamics, plasma physics, and cosmological modeling are now beginning to explore the integration of PINNs, suggesting that this framework may yet evolve into a staple of computational physics [48]. However, widespread adoption will likely require deeper theoretical guarantees, faster training methods, and improved interpretability in the face of domain-specific complexity.

## 4 Challenges, Limitations, and Theoretical Gaps

While the conceptual elegance and empirical utility of Physics-Informed Neural Networks (PINNs) have been repeatedly demonstrated across a diverse spectrum of applications, their deployment remains fraught with multifaceted challenges

that stem not merely from algorithmic inefficiencies, but from deeper issues of mathematical ill-posedness, optimization instability, and theoretical incompleteness. In this section, we undertake a meticulous dissection of these limitations, organizing them into categories that reflect their origin: optimization dynamics and expressivity, loss landscape pathology and stiffness, conditioning and generalization, as well as foundational theoretical deficits that still pervade the literature [49].

#### 4.1 Optimization Landscape and Training Dynamics

At the heart of every PINN implementation lies an optimization procedure tasked with minimizing a composite loss functional  $\mathcal{L}_{\text{PINN}}(\theta)$ , which incorporates both data-driven and physics-derived terms [50]. The empirical convergence of this objective function is, however, notoriously sensitive to the choice of architecture, initialization, and optimizer dynamics. In stark contrast to conventional supervised learning tasks, the loss landscape in PINNs is shaped by high-order differential operators whose gradients can introduce numerical instabilities, nonlocal dependencies, and ill-conditioning. These effects are exacerbated when the underlying physical system is governed by stiff or chaotic PDEs, in which case vanishing gradients and erratic curvature render gradient descent ineffectual [51]. Furthermore, the balance of the different components of the PINN loss function—e.g., between PDE residuals, initial conditions, and observational data—remains more art than science [52]. The manual selection or heuristic tuning of the weighting coefficients  $\{\lambda_{\text{data}}, \lambda_{\text{res}}, \lambda_{\text{IC}}, \lambda_{\text{BC}}\}$  introduces subjectivity and may inadvertently bias the model toward overfitting one component at the expense of another. While various adaptive weighting schemes and dynamic loss balancing heuristics have been proposed, none offer universal guarantees, and the underlying theoretical principles guiding such heuristics are poorly understood [53].

#### 4.2 Spectral Bias, Stiffness, and Function Space Coverage

Neural networks trained via standard gradient-based methods are known to exhibit a spectral bias: they tend to learn low-frequency components of the target function earlier than high-frequency ones. This inherent preference becomes problematic in physical systems characterized by sharp gradients, boundary layers, or multiscale behavior—conditions under which high-frequency components are essential [54]. In such contexts, the training process stagnates or fails to recover crucial solution features [55]. This phenomenon, termed the *frequency principle* or *spectral bias*, has been observed empirically in PINNs and motivates the use of Fourier-feature embeddings, sinusoidal activation functions (SIRENs), and hierarchical multi-scale architectures as countermeasures. Closely related to this is the issue of stiffness, particularly prevalent in equations with disparate timescales, such as reaction-diffusion systems, combustion dynamics, or stiff chemical kinetics [56]. Here, the rapid variation of solution trajectories

demands highly expressive and numerically stable approximators [57]. Unfortunately, standard PINNs lack intrinsic mechanisms to handle such stiffness, often requiring impractically high collocation point densities or unsustainable training times [58]. Attempts to alleviate this include domain decomposition (XPINNs), operator splitting, and the use of neural ordinary differential equation (ODE) solvers; however, a universally robust solution remains elusive.

### 4.3 Generalization, Overfitting, and Data Efficiency

Despite the name “physics-informed,” PINNs are not immune to classical pitfalls of machine learning, including overfitting, poor generalization, and data inefficiency. The imposition of physics constraints via residual losses does not inherently guarantee correct generalization outside the training domain, particularly when the governing PDE is only partially known or inaccurately specified [59]. Moreover, in inverse problems, where the parameter space can be vast or unconstrained, PINNs may converge to spurious local minima that satisfy the PDE residuals but are physically nonsensical [60]. The promise of data efficiency—central to the PINN ethos—is not uniformly realized across problem settings. In high-dimensional PDEs or inverse problems with multiple latent parameters, the number of collocation and observation points needed to maintain solution accuracy can grow exponentially, counteracting any notional advantage over classical methods [61]. Recent hybrid schemes combining PINNs with surrogate models or reduced-order methods (e.g., Proper Orthogonal Decomposition, POD; or Dynamic Mode Decomposition, DMD) have shown potential in improving sample efficiency, though their integration remains an open research area.

### 4.4 Boundary Condition Enforcement and Geometric Complexity

The proper enforcement of boundary conditions remains a persistent and often underappreciated challenge in the PINN framework. In geometrically complex or non-convex domains, the naive sampling of collocation points may fail to adequately cover the boundary, leading to weak enforcement and degraded solution fidelity [62]. While “hard” constraint methods—wherein the network architecture is tailored to exactly satisfy boundary conditions—offer one remedy, their implementation becomes nontrivial in higher dimensions or for non-rectilinear geometries [63]. Furthermore, the encoding of boundary conditions into the loss function via penalty terms introduces additional hyperparameters and may not ensure exact satisfaction [64]. Recent proposals, including conformal mappings, coordinate transformations, and level-set-based methods, seek to mitigate this issue but come at the cost of architectural complexity and potential loss of generality [65].

### 4.5 Lack of Theoretical Guarantees and Convergence Analysis

Perhaps the most significant limitation of PINNs is the lack of comprehensive theoretical understanding that can explain, predict, or guarantee their convergence behavior. Classical numerical methods are undergirded by well-developed

theories of stability, consistency, and convergence; by contrast, PINNs lack rigorous analogs for most of these constructs [66]. Although universal approximation theorems guarantee that neural networks can represent solutions to arbitrary accuracy under certain conditions, these results are existential in nature and do not provide constructive guidance for practical network design or training procedures [67]. Only limited results are available regarding the convergence of PINN approximations in Sobolev norms, and these are typically confined to linear or weakly nonlinear PDEs under strong regularity assumptions [68]. Moreover, there exists no general theory for the error propagation in PINNs, particularly when both measurement noise and PDE model error are present [69]. As such, current PINN implementations remain largely heuristic, relying on empirical validation rather than principled analysis [70].

#### 4.6 Computational Cost and Scalability

The training of PINNs is computationally intensive, often requiring thousands to millions of iterations of high-dimensional optimization. Unlike traditional solvers whose computational complexity scales predictably with mesh size, PINNs incur substantial overhead due to the need for backpropagation through differential operators and the training of deep architectures [71, 72]. This inefficiency is exacerbated in multidimensional or multiphysics settings, where the evaluation of PDE residuals may involve coupled or nested differential equations [73].

Parallelization strategies, GPU acceleration, and mini-batching techniques offer partial mitigation, but scalability to industrial-scale problems remains a formidable hurdle [75]. Novel formulations such as operator-learning PINNs, physics-informed transformers, and mesh-adaptive PINNs are active areas of research aimed at reducing computational costs without sacrificing accuracy or physical fidelity.

#### 4.7 Summary of Limitations

In summary, the limitations of PINNs are both deep and wide-ranging, encompassing mathematical, algorithmic, and practical dimensions [76]. Their failure modes are often subtle and problem-dependent, requiring a nuanced understanding of both the physical system at hand and the peculiarities of machine learning optimization. Addressing these challenges demands a cross-disciplinary effort, combining insights from numerical analysis, deep learning theory, differential geometry, and domain-specific physics.

### 5 Future Research Directions and Open Questions

As the intellectual and technological edifice of Physics-Informed Neural Networks (PINNs) continues to evolve, the frontier of this research landscape reveals a multitude of intricate and interdependent challenges, the resolution of which may fundamentally redefine not only the scope of data-driven scientific

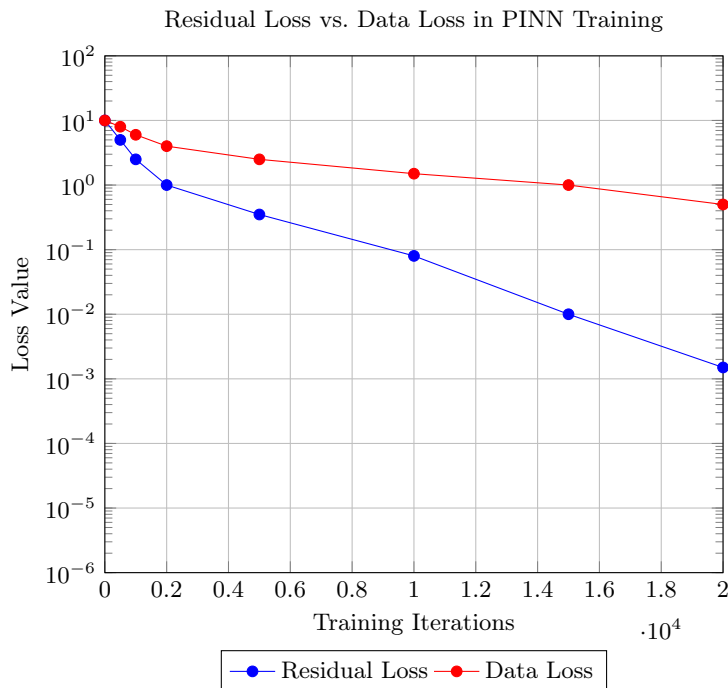


Fig. 3: Comparison of residual and data loss during the training process of a PINN. Both loss components decrease over time, with the residual loss (blue) representing the PDE-based constraint and the data loss (red) minimizing the discrepancy between model predictions and observed data [74]. The logarithmic scale highlights the relative rate of convergence of both components.

computing but also our epistemological approach to modeling physical systems [72]. In this concluding section, we articulate a selection of prominent research directions and formulate several pressing open questions that, collectively, delineate the contours of a comprehensive research agenda for the PINN community. Our discussion is framed along the axes of theoretical development, algorithmic innovation, architectural advancement, and interdisciplinary integration.

### 5.1 Toward a Rigorous Mathematical Theory of PINNs

At present, the mathematical foundations of PINNs remain largely heuristic, lacking in robust error estimates, convergence guarantees, and approximation theory tailored to physics-constrained learning [77]. A key priority for future work is the derivation of bounds on the generalization error  $\mathcal{E}_{\text{gen}}$  in the presence of noisy observations, imperfect PDE models, or incomplete boundary data. Such bounds must necessarily account for the interplay between network capacity, sample complexity, and the regularity of the underlying PDE solution

in appropriate Sobolev spaces [78]. Of particular interest is the characterization of expressivity and approximation power of neural networks under physics-informed constraints [79]. While classical results (e.g., Barron-type theorems) describe neural network approximation in terms of  $L^2$  norms, the need for high-order derivative accuracy in PDEs demands new theoretical constructs that operate in  $H^k$  or weighted Sobolev norms. Questions remain open regarding the compactness of the function class induced by PINNs and whether such classes admit universal approximation properties in domains with complex topologies or singularities [80].

## 5.2 Robust and Scalable Optimization Methods

Another critical bottleneck in PINN deployment is the lack of optimization algorithms specifically designed for PDE-constrained loss landscapes. Future work must aim to go beyond traditional stochastic gradient descent (SGD) and its momentum-based variants, developing solvers that incorporate curvature information, operator-adapted preconditioners, or adaptive collocation strategies informed by residual variance [81]. A promising direction involves the development of hybrid optimization frameworks that alternate between solving physics-constrained least squares subproblems and updating network weights in a constrained fashion. Such methods might draw inspiration from sequential quadratic programming (SQP), augmented Lagrangian formulations, or interior-point methods tailored to the nonconvex structure of neural network training. Moreover, the use of surrogate models to approximate costly PDE evaluations in the loss functional, while maintaining fidelity, could dramatically reduce computational burdens [82].

## 5.3 Architectural Innovations and Operator Learning

While conventional PINNs employ feedforward neural networks with static depth and width, future architectures will likely require greater adaptability and physical interpretability. One emerging paradigm is the use of neural operators—such as DeepONets, Fourier Neural Operators (FNOs), or attention-based transformers—that learn mappings between function spaces, thus generalizing the PINN philosophy from equation-specific solutions to operator-level emulation [83]. The integration of geometric deep learning, including graph neural networks (GNNs) and differential manifold embeddings, represents another frontier. These architectures may offer natural representations for physical systems with non-Euclidean structure, such as curved surfaces, topological defects, or spatially adaptive meshes. The exploration of equivariant and invariant networks, which respect symmetries of the governing equations (e.g., Galilean, Lorentz, or gauge invariance), could further enhance model generality and sample efficiency [84].

#### 5.4 Data Assimilation, Uncertainty Quantification, and Bayesian PINNs

A pressing need exists to equip PINNs with principled mechanisms for data assimilation and uncertainty quantification (UQ), particularly in settings characterized by noisy or incomplete measurements [85]. Bayesian variants of PINNs (B-PINNs), which treat weights, inputs, or even PDE parameters as random variables with posterior distributions conditioned on data and physical constraints, offer a fertile avenue for exploration. However, scalable and accurate posterior inference in such high-dimensional spaces remains a formidable challenge. Future research may benefit from advances in probabilistic programming, variational inference, and Hamiltonian Monte Carlo methods to render Bayesian PINNs tractable in practice [86]. Moreover, the integration of UQ with adaptive sampling—wherein new observations are acquired to minimize predictive entropy or maximize information gain—could lead to efficient experimental design strategies grounded in physical consistency [87].

#### 5.5 Hybrid Modeling and Scientific Co-Design

In many real-world applications, neither purely mechanistic models nor purely data-driven models suffice [88]. Hybrid modeling frameworks that integrate partial knowledge from first principles with data-driven components—whether via residual learning, symbolic regression, or multi-fidelity modeling—represent an increasingly important direction [89]. PINNs naturally lend themselves to such hybridization, but systematic methodologies for coupling them with legacy solvers, surrogate models, or even symbolic dynamical systems remain underdeveloped. Furthermore, the vision of scientific machine learning as an end-to-end co-design paradigm—where experiment, simulation, and inference co-evolve—calls for PINNs to be embedded within larger feedback loops [90]. These loops might include sensor placement, experiment control, and hypothesis testing, thereby shifting the role of PINNs from passive approximators to active scientific agents.

#### 5.6 Inverse Problems and Hidden Physics Discovery

One of the most tantalizing promises of PINNs lies in their ability to infer unknown parameters, source terms, or even structural forms of governing equations from sparse observations [91]. However, the reliability of such inference remains deeply sensitive to identifiability, regularization, and prior assumptions [92]. Future work must rigorously define identifiability conditions under which a given class of PDEs, boundary conditions, and measurements admits a unique solution for the unknowns of interest [93]. Symbolic regression, sparse identification (e.g., SINDy), and neural architecture search may be fused with PINNs to uncover latent physical laws in a form that is both interpretable and computationally efficient [94]. At the same time, care must be taken to distinguish spurious correlations from genuine causal mechanisms, particularly in noisy or over-parameterized regimes [95].

### 5.7 Open Problems and Grand Challenges

Several foundational open questions remain unresolved and merit long-term attention:

- Under what conditions do PINNs converge to the true solution of a PDE as the number of collocation points increases and the network capacity is scaled?
- Can one define a notion of stability and consistency for PINNs analogous to that in finite element or finite difference methods [96]?
- What are the necessary and sufficient conditions for the identifiability of unknown physics in inverse PINN settings?
- How can PINNs be rendered interpretable and verifiable for deployment in safety-critical or regulatory environments?
- What is the information-theoretic limit of learning physics from data under noise, sparsity, and model uncertainty [97]?

These questions point not only to limitations but also to the tremendous potential of PINNs as a unifying language for scientific inference, modeling, and simulation. Addressing them will require a concerted and interdisciplinary effort, drawing upon insights from applied mathematics, statistical physics, optimization theory, machine learning, and beyond [98].

## 6 Conclusion

In synthesizing the rapidly evolving domain of Physics-Informed Neural Networks (PINNs), we have sought to elucidate the underlying mathematical principles, architectural formulations, and algorithmic mechanisms that collectively define this emergent paradigm at the intersection of machine learning and scientific computing. By embedding the governing laws of physics—typically expressed as systems of partial differential equations—directly into the loss functional of neural network training objectives, PINNs represent a fundamentally novel approach to the data-driven approximation of dynamical systems [99]. This survey has endeavored to offer a comprehensive and critical account of the techniques, applications, trends, and theoretical challenges that characterize this field [100]. The successes of PINNs across diverse applications—from fluid mechanics and materials science to climate modeling and biophysics—are as impressive as they are varied, offering empirical validation of the approach’s versatility [101]. Nonetheless, our analysis has demonstrated that such successes often mask deep structural limitations: training instability due to stiff PDEs, spectral bias limiting high-frequency approximation, boundary enforcement inconsistencies, poor scalability, and a conspicuous absence of rigorous convergence theory. While promising remedies have emerged in the form of adaptive sampling, hybrid architectures, multi-fidelity frameworks, and operator learning paradigms, these innovations remain largely heuristic and are yet to be unified under a rigorous mathematical framework.

We have argued that the future maturation of PINNs as a viable and trustworthy scientific tool necessitates advances on several fronts: (1) the development of foundational theoretical frameworks analogous to those in classical numerical analysis, including notions of stability, convergence, and consistency; (2) the design of new optimization strategies and loss formulations tailored to the peculiarities of PDE-constrained learning; (3) the creation of architectures that encode geometric and physical inductive biases in a principled and scalable manner; and (4) the integration of uncertainty quantification and data assimilation into the PINN framework to enable robust inference in real-world scenarios with noisy, sparse, or incomplete data.

Moreover, we have highlighted the importance of interdisciplinary collaboration—between physicists, applied mathematicians, computer scientists, and engineers—in advancing the field. PINNs do not merely represent a new numerical technique but rather a paradigm shift in how we conceive of modeling, learning, and inference in physical systems. They invite a reconceptualization of the boundary between model-driven and data-driven approaches, proposing a synthesis that is simultaneously informed by domain knowledge and flexible enough to accommodate empirical complexities.

As the field advances, the challenge will be not only to extend the range of problems for which PINNs are applicable but also to elevate the level of understanding to the point where their behavior is predictable, their limitations are quantifiable, and their outcomes are reliable. It is through this lens that we must approach future work—with both ambition and epistemic humility—recognizing that the true integration of physics and learning remains a profound and open scientific quest.

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